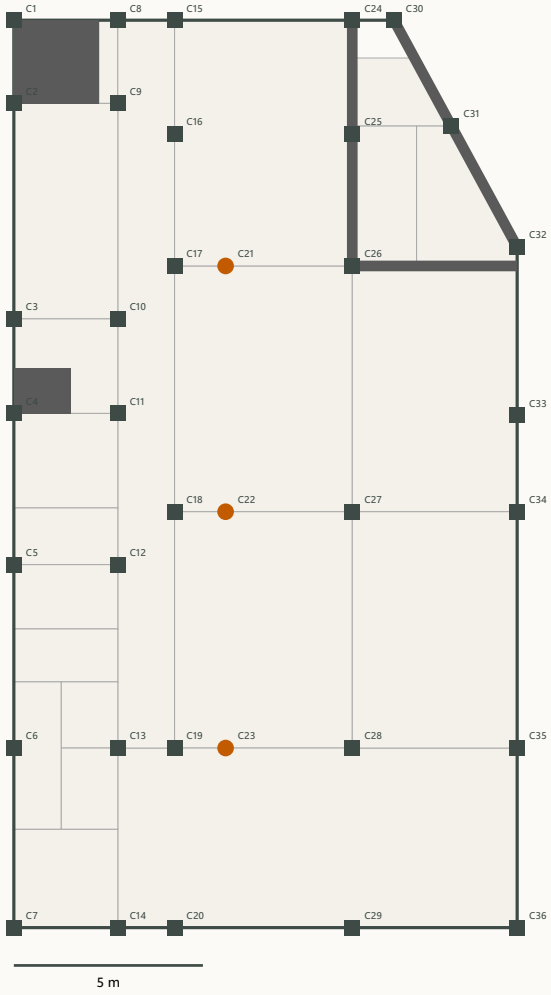
Preliminary structural study · Taif / Palm House V6.2

architectural coordination study · every size ASSUMED, to be calculated by the structural engineer · CSV coordinates, north up

Ground floor · columns (■ to the roof, ● colonnade to the FF), RC boxes (dark), tie beams on the column lines



The idea: a simple RC skeleton, walls on walls

- A **reinforced-concrete frame** (columns, beams, slabs) with **block infill**, as Saudi contractors build every villa. Columns stand on the existing wall lines, hidden in the walls; **every column runs from its footing to the roof** (the three colonnade columns end at the first floor, nothing starts on them except the edge beam). No transfer beam carries a column.
- **Spans 2.2–6.5 m.** The only spans above 6 m are on the courtyard lines (x 4.25, 5.6, 8.95: 6.5 and 6.25 m), because the courtyard and the double-height living must stay free of columns; they get drop beams ≈ 0.50–0.55 deep.
- **Slabs:** solid 0.20 m where spans are small (west band, riwaq, projection, roofs of 4.7 m), hollow-block **ribbed 0.30 m** (25+5) for the 4.36–4.75 m bays of the east band and the men's majlis roof. The double-height living's roof is a **flat solid slab 0.22 m spanning 4.7 m east–west** between the two long walls: **no beam crosses the void** (§ options).
- **Foundations (assumed):** isolated pad footings ≈ 1.5 × 1.5 m under the columns, strip footings under the RC boxes, RC tie beams on the column lines at ground level; the lift pit 1.2 m deep; the air tunnel under the east band crosses the tie beams (depth to coordinate). **No soil report yet** (X9).

Balanced stiffness (no twist)

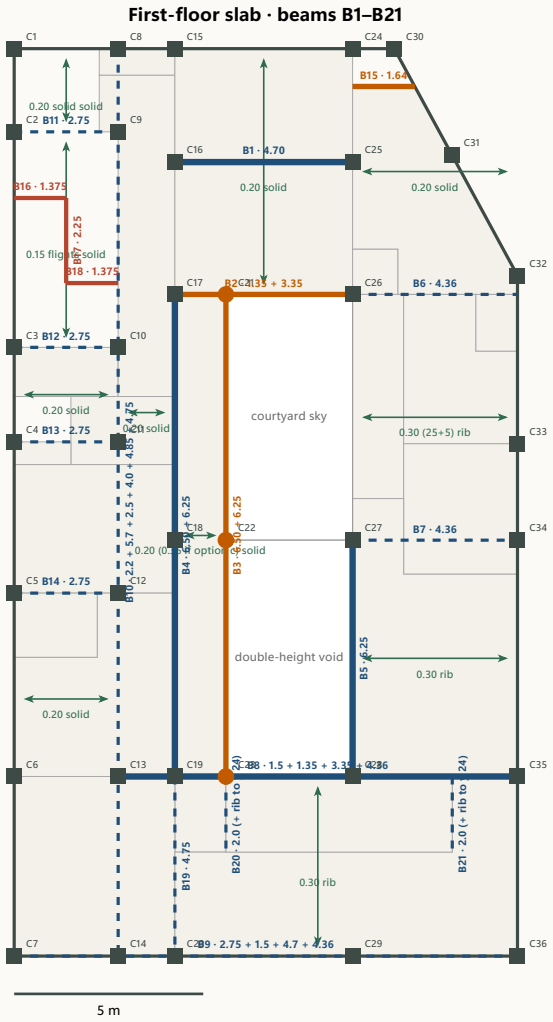
- **West:** the lift shaft (W1, RC walls on 3 sides), the west shaft (W2) and the stair walls make a stiff core at x 0–2.75.
- **East, to match it:** the solid laundry / maid wall on x 8.95 (W3, no openings, 6.5 m), the cut wall (W4) and the kitchen's north wall (W5) are built as RC walls 0.20 m: an east box of the same order of stiffness, at a similar distance from the centre. With the long north and south walls (block infill in the frame) the centre of stiffness stays near the centre of mass.
- **Why:** the house is a long pentagon (13.3 × 24 m); with stiff elements only on the west, wind and earthquake (Taif is a low seismic zone, but the SBC still asks for it) would twist it about the west core.

Column lines and spans

Line	Columns at y (or x)	Spans (m)
x 0 and x 2.75	y 0 · 2.2 · 7.9 · 10.4 · 14.4 · 19.25 · 24	2.2 · 5.7 · 2.5 · 4.0 · 4.85 · 4.75
x 4.25 and x 8.95	y 0 · 3.0 · 6.5 · 13.0 · 19.25 · 24	3.0 · 3.5 · 6.5 · 6.25 · 4.75
x 5.6 (colonnade, to the FF)	y 6.5 · 13.0 · 19.25	6.5 · 6.25
cut wall	(10.051, 0) · (11.572, 2.8) · (13.311, 6.0)	3.2 · 3.6
x 13.311	y 6.0 · 10.45 · 13.0 · 19.25 · 24	4.45 · 2.55 · 6.25 · 4.75
y 0 / y 24 (cross)	x 0 · 2.75 · 4.25 · 8.95 · 13.311 (cut corner at 10.051)	2.75 · 1.5 · 4.7 · 4.36 (1.1)
y 6.5 · 13.0 · 19.25 (cross)	x 4.25 · 5.6 · 8.95 · 13.311 (+ 2.75 at y 19.25)	1.35 · 3.35 · 4.36

First-floor slab (+3.40 … +3.60) · beams and slab directions

blue = drop beams · dashed = hidden beams (slab depth) · orange = edge beams of the openings · red = stair trimmers · green arrows = slab span · sizes width × depth, assumed



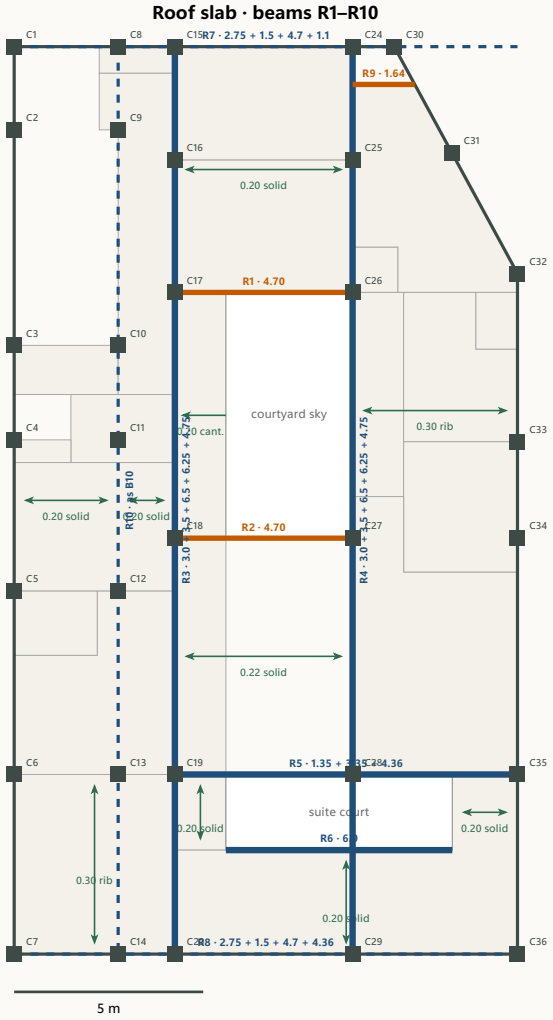
Beam	Span (m)	Size, assumed	Carries / note
B1	4.70	0.25×0.40 drop (or hidden in a 0.30 ribbed slab)	FF kids' room / family living wall (y 3.0, nothing below) + the women's majlis roof
B2	1.35 + 3.35	0.25×0.40 hidden / drop	FF slab edge on the courtyard (family living balustrade); over the women's majlis south glazing
B3	6.50 + 6.25	0.25×0.50 · drop, UPTURNED or 0.35 slab edge (§ options)	the 1.35 m projection and the gallery, on C21–C23
B4	6.50 + 6.25	0.25×0.50 drop	FF west rooms' east wall (over the riwaq glazing and the low wall) + the projection back span
B5	6.25	0.25×0.55 drop (or 0.60×0.40 wide-shallow)	FF suite bedroom west wall + half the dining roof; over the open ceiling-height change
B6	4.36	hidden 0.30	on the kitchen north wall; FF bedroom 1 / suite wall above
B7	4.36	hidden 0.30	on the kitchen / dining wall; the suite dressing floor spans over it
B8	1.5 + 1.35 + 3.35 + 4.36	0.25×0.45 drop above the wall	on the men's majlis north wall; carries the FF bedroom 3 / office wall, the suite court north wall, the 7 m living south wall and the treadmill arm
B9	2.75 + 1.5 + 4.7 + 4.36	hidden in the 0.25 wall / 0.25×0.45	south wall; men's majlis roof support
B10	2.2 + 5.7 + 2.5 + 4.0 + 4.85 + 4.75	hidden 0.20–0.30; 0.25×0.40 at the 5.7 m stair bay	riwaq line: west band slabs and the riwaq slab
B11	2.75	hidden 0.20	lift / stair wall
B12	2.75	hidden 0.20	stair / utility wall · FF landing
B13	2.75	hidden 0.20	utility / hospitality kitchen wall
B14	2.75	hidden 0.20	basins / family WC wall · FF bedroom 2 / bath 3 wall
B15	1.64	0.20×0.35 edge	air tower opening (the tower rises through both slabs)
B16	1.375	0.20×0.40 trimmer	stair opening; spans to the x 1.375 trimmer (a beam on a beam: unavoidable at a stair well)
B17	2.25	0.20×0.40 trimmer	stair opening, between the two flights
B18	1.375	0.20×0.40 trimmer	stair opening, arrival edge
B19	4.75	hidden rib 0.30 (in the ribbed majlis roof)	FF office / gym route wall, nothing below; a thickened rib spanning y 19.25 → 24
B20	2.0 (+ rib to y 24)	hidden rib 0.30	FF gym route / suite court wall, nothing below
B21	2.0 (+ rib to y 24)	hidden rib 0.30	the treadmill arm's west wall, nothing below; treadmill live + dynamic load

First-floor walls and what carries them

On a wall below (block infill under a hidden beam)	Nothing below → beam or light partition
x 0, x 13.311, y 0, y 24, the cut wall (outer walls); x 2.75 at the lift and landing; x 2.25 lift; y 2.2, 7.9, 10.4 (shaft), 14.4, 16.1 (bath 3 = GF family WC), 19.25 (majlis north wall), x 1.5 shaft; x 8.95 y 6.5–13 (kitchen west wall); y 6.5 x 8.95–13.311 (kitchen north wall); y 13.0 line inside the suite (kitchen / dining wall)	**y 3.0** kids' / family living → **B1**; **x 4.25 y 13–19.25** bedrooms' east wall over the low wall / opening → **B4**; **x 8.95 y 13–19.25** suite bedroom west wall over the open ceiling change → **B5**; **x 4.25 y 19.25–24** office → **B19** rib; **x 5.6** gym route → **B20**; **x 11.586** treadmill arm → **B21**; **y 21.25** gym north wall (7.7 m, east–west across the ribs) → **light partition on the slab** (X6); suite walls x 10.3, y 10.45, x 12.211, y 8.0, y 11.9, y 13.9 over the kitchen → light partitions (100 mm block or dry wall) on the ribbed slab, or hidden beams under the bath walls; y 11.0 (bath / niche / bedroom 2) and y 9.2 x 1.5–4.25 (family bath north) → light partitions; x 2.2 bath 3 / bedroom 3 entry → light; y 5.3 and x 10.15 (bedroom 1 / family link) → light; linen front y 0.7 → joinery

Roof slab (+7.00 … +7.20) · beams and slab directions

the courtyard sky and the suite court are holes; the double-height living is roofed by a flat slab with no beam in the void; the gallery strip cantilevers (or C21–C22 continue to the roof)



Beam	Span (m)	Size, assumed	Carries / note
R1	4.70	0.25×0.45 drop	courtyard sky hole, north edge (over the FF family living balustrade line)
R2	4.70	0.25×0.45 drop	courtyard sky hole, south edge = the clean double-height roof's north edge
R3	3.0 + 3.5 + 6.5 + 6.25 + 4.75	0.25×0.50 drop above the FF wall	on the FF west rooms' east wall / office wall; the roof over the gallery strip cantilevers 1.35 m from it (or C21–C23 to the roof)
R4	3.0 + 3.5 + 6.5 + 6.25 + 4.75	0.25×0.50 drop above the FF wall	on the FF suite west wall; the double-height roof spans to it
R5	1.35 + 3.35 + 4.36	0.25×0.50 drop	over the living's south wall (clerestory head +6.00, soffit ≥ +6.50) and the suite court north wall
R6	6.0	0.25×0.50 DROP (not upturned: winter light, X4)	suite court hole, south edge = gym roof north edge; on the FF walls x 5.6 and x 11.586 (a transfer, X5)
R7	2.75 + 1.5 + 4.7 + 1.1	hidden in the 0.25 wall	north wall
R8	2.75 + 1.5 + 4.7 + 4.36	hidden in the 0.25 wall	south wall
R9	1.64	0.20×0.35 edge	air tower head opening
R10	as B10	hidden 0.20–0.30	riwaq line

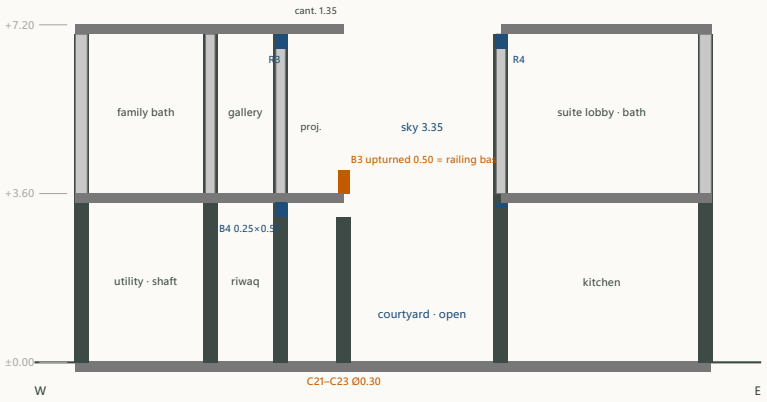
Slab bays (first floor and roof)

Level	Bay (x0 y0 – x1 y1)	Type	Span	Note
FF	0 0 – 2.75 2.2	lift: FF slab panel, removable later 0.20 solid	NS	future lift
FF	0 2.2 – 2.75 7.9	stair: L-shaped opening + flight slabs 0.15 flights	NS	opening x 0–2.75 y 2.2–3.95 + x 1.375–2.75 y 3.95–6.2
FF	0 7.9 – 2.75 10.4	solid 0.20	EW	utility; shaft opening 1.5×1.2 framed
FF	0 10.4 – 2.75 24	solid 0.20	EW	west wash block, 2.75 m bays
FF	2.75 0 – 4.25 19.25	solid 0.20	EW	riwaq, 1.5 m
FF	4.25 0 – 8.95 6.5	solid 0.20	NS	women's majlis roof: 3.0 + 3.5 m to beam B1
FF	4.25 6.5 – 5.6 19.25	solid, the projection 0.20 (0.35 if option c)	EW	1.35 m between B4 and the edge beam B3
FF	8.95 0 – 13.31 6.5	solid 0.20	EW	maid wing, 4.36 m to the cut wall
FF	8.95 6.5 – 13.31 13	ribbed (hollow block) 0.30 (25+5)	EW	kitchen roof; suite bath sunken 0.15 (depressed zone x 10.3–13.311 y 6.5–10.45)
FF	8.95 13 – 13.31 19.25	ribbed 0.30	EW	dining roof; half on beam B5
FF	2.75 19.25 – 13.31 24	ribbed 0.30	NS	men's majlis roof, 4.75 m; thickened ribs B19–B21 under the FF walls; gym live load
roof	0 0 – 2.75 24	solid 0.20	EW	west band and lift / shaft heads
roof	2.75 0 – 4.25 24	solid 0.20	EW	riwaq / gallery line
roof	4.25 0 – 8.95 6.5	solid 0.20	EW	kids' room / family living: 4.7 m, no beam on y 3.0 (light partition)
roof	4.25 6.5 – 5.6 13	solid, cantilever 0.20	cant W	over the gallery strip: 1.35 m cantilever from R3 (or on C21–C22 extended)
roof	4.25 13 – 8.95 19.25	solid, flat, no beams in the void 0.22	EW	the double-height living roof: 4.7 m R3 → R4, clean soffit
roof	8.95 0 – 13.31 19.25	ribbed 0.30	EW	bedroom 1 / suite: 4.36 m
roof	0 19.25 – 4.25 24	ribbed 0.30	NS	office roof: 4.75 m
roof	4.25 19.25 – 5.6 21.25	solid 0.20	NS	gym route
roof	4.25 21.25 – 13.31 24	solid 0.20	NS	gym roof: 2.75 m; its north edge R6 over the court
roof	11.59 19.25 – 13.31 21.25	solid 0.20	EW	treadmill arm roof

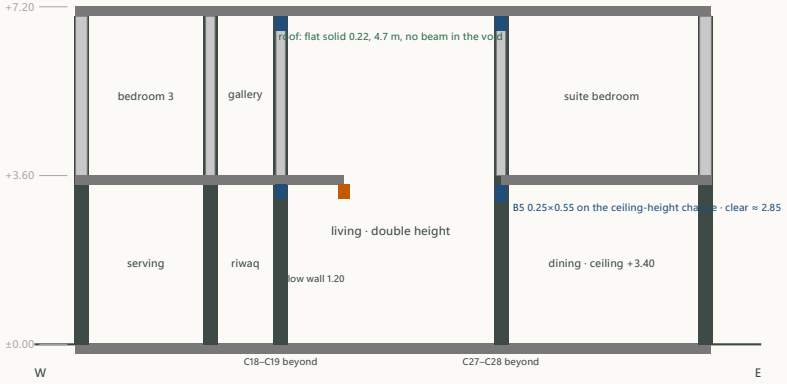
Sections · schematic, levels to confirm

A east–west through the riwaq, the courtyard and the colonnade (y 10) · B east–west through the living, the double-height void and the dining (y 16.5) · C north–south through the middle band (x 7.3)

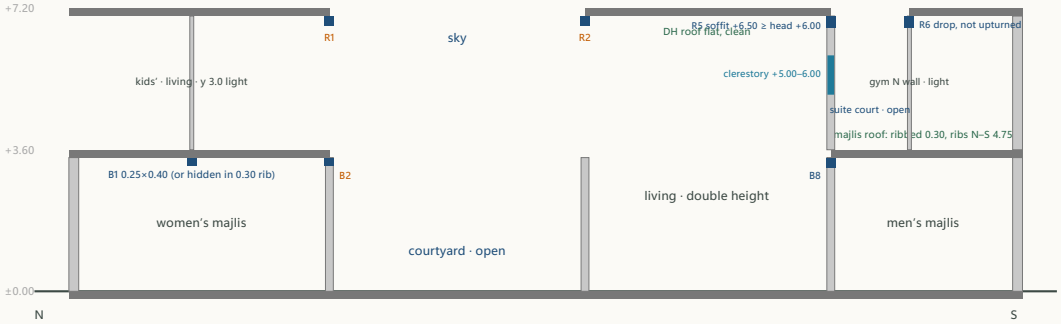
A · y 10: riwaq · courtyard · colonnade · kitchen



B · y 16.5: living (low wall) · double-height void · dining · suite



C · x 7.3: women's majlis · courtyard · double-height living · men's majlis roof · gym



Special zones · options compared (none imposed)

1 · The west projection: edge beam B3 on C21–C23 (spans 6.5 and 6.25 m)

	(a) drop beam	(b) upturned beam	(c) deeper slab edge
size	0.25 × 0.50 below the slab	0.25 × 0.50 above the slab, top at +4.10	slab 0.35 over the 1.35 m strip, no beam
what you see	a 0.30 m downstand along the colonnade, clear 2.90 under it; the courtyard reads as a riwaq with a beam	a flush soffit at +3.40 over the colonnade; the upstand is the solid base of the gallery railing (0.50 + 0.60 railing = 1.10)	flush soffit, a thicker slab edge 0.35 seen from the courtyard
structure	standard; the stiffest	standard; the slab hangs from the beam's bottom (stirrups) – routine	a 1.35 m wide, 0.35 deep band over 6.5 m: depth/span ≈ 1/18, deflection-sensitive, heavy
cost	lowest	≈ same formwork; saves the railing base	more concrete and steel
note	—	preferred by this study: cleanest colonnade soffit, railing base included	only if a flush soffit AND an open railing are wanted

2 · The double-height living: a clean ceiling (preferred, not a rule)

- **Option 1 · flat solid roof slab 0.22 m spanning 4.7 m east–west** from the FF west rooms' wall (R3) to the suite wall (R4): no beam in the void, the two beams sit on the walls (invisible). Routine span; cost ≈ a normal solid slab. **This study's preference.**
- **Option 2 · ribbed 0.30 slab, ribs east–west:** lighter and cheaper for the contractor, but a ribbed soffit (or a false ceiling) over the void; still no cross beam.
- **Option 3 · a cross beam on y 16.1 (mid-void) + 0.18 slab:** the cheapest in material, but a 0.45 m beam across the void, 7 m up, in the family's main room. Not recommended unless the engineer finds option 1 too deep.
- The void's walls: x 4.25 and x 8.95 are 7 m high but tied by the FF slab on their outer sides (the gallery / projection on the west, the suite floor on the east). The south wall (y 19.25, x 5.6–8.95, with the clerestory) is tied at +3.6 only by the men's majlis roof slab on its south face, and at +7.0 by the roof: **one tall wall, 7 m, tied on one side at mid-height** (X7).

3 · Living ↔ dining opening (x 8.95, 6.25 m) under the suite wall: beam B5

- A drop beam 0.25 × 0.55 between C27 (8.95, 13.0) and C28 (8.95, 19.25); it carries the suite bedroom's west wall (3.4 m block), half the dining roof (ribbed, 4.36 m) and the roof beam R4 above the wall. Soffit ≈ +2.85.
- **It can sit exactly on the ceiling-height change line:** the dining ceiling is at +3.40, the living has no ceiling, so the beam's 0.35 m below the dining ceiling becomes the step itself: one face of the double height. Headroom under it 2.85 m (≥ 2.40 required).
- **Alternative:** a wide-shallow beam 0.60 × 0.40 (soffit +3.00) if the step should be thinner; or make the suite wall a light partition so the beam carries only the slab (0.25 × 0.45).

4 · Men's majlis roof (10.56 × 4.75 m) carrying the gym, the arm and the office

- **Span direction north–south, 4.75 m:** a ribbed (hollow-block) slab 0.30 m (25+5), ribs from the majlis north wall (B8) to the south wall (B9). **No column inside the majlis** (the room must stay free: 10.56 m long, seating at the east end).
- The first-floor walls **parallel to the ribs** (office wall x 4.25, gym route x 5.6, treadmill arm x 11.586) each stand on a **thickened rib / hidden beam** B19–B21 spanning the 4.75 m: direct and cheap.
- The gym's north wall **y 21.25 runs across the ribs** for 7.7 m with nothing below: (i) **a light partition** (100 mm block or dry wall, high windows onto the court) on the slab, designed for a partition load: **recommended**; (ii) a 200 mm block wall on a solid band 1.0 m wide in the ribbed slab; (iii) a drop beam the full 10.56 m (≈ 0.90 m deep, visible in the majlis): rejected.
- **Loads:** gym live load 5 kN/m² (SBC 301 / ASCE 7: gymnasia), the treadmill ≈ 1.5–2 kN dynamic; office 2.5–3 kN/m². **Vibration:** a 4.75 m ribbed slab is stiff (natural frequency well above 8 Hz expected), the treadmill arm gets a floating floor and pads (DETAILS.md); the engineer to check the frequency and the slab's response under the treadmill (X6).

5 · The stair (x 0–2.75, y 2.2–7.9)

- Two flights side by side, 1.375 m wide each (assumed), 22 risers × 0.164 = 3.60 m, goings 0.28: flight 1 rises from y 7.9 north to a half landing (+1.80, y 2.2–3.4, the store below it, ≈ 1.6 m headroom); flight 2 comes back south and **arrives at y ≈ 6.2 at the south end**, onto the FF landing (y 7.9–9.2) through the stair box's arrival strip.
- **Opening in the FF slab (L-shaped):** over the half landing and the top of flight 1 (x 0–2.75, y 2.2–3.95: headroom ≥ 2.10 is lost where the floor is above +1.30) and over flight 2 (x 1.375–2.75, y 3.95–6.2). ≈ 8.0 m².
- **Framing:** trimmers B16 (y 3.95), B17 (x 1.375, between the flights) and B18 (y 6.2), 0.20 × 0.40, on the stair walls x 0 / x 2.75 (columns C2, C3, C9, C10); B16 and B18 bear on B17: the one beam-on-beam of the house, normal at a stair well. Flights: solid slabs 0.15 spanning landing to landing (2.8 m).
- **Headroom:** under the FF slab over flight 1's lower part ≥ 2.10 (checked by the opening's extent); under the roof over flight 2 ≥ 3.0.

6 · Glazing and the grid

- **Courtyard glazing** (x 4.25 y 6.8–8.9 and 10.7–12.7; x 8.95 y 9.1–11.1): no column in the openings (C17, C18, C26, C27 at the corners); beams B4 / B6 above the heads. **Women's majlis south glazing** (y 6.5, x 5.1–8.1): C21 at x 5.6 sits in it → the glazing is 5.1–5.4 + 5.8–8.1 (accepted with the colonnade). **Living glazing** (y 13.0, x 4.45–8.75): C22 at x 5.6 is the pier (the columns solution, tag v5.0). **Clerestory** (y 19.25, x 6.0–7.0 and 7.6–8.6, +5.00–6.00): C23 at x 5.6 and C28 at x 8.95 are clear of it; the roof beam R5 soffit +6.50 is above the head. **Suite bath east window** (y 9.6–10.35): C33 at y 10.45 clashes (X3).

7 · Lift shaft, air tower, roof openings

- **Lift (future, two stops):** RC walls W1 on x 0, y 0 (outer) and y 2.2, 0.20 m; the east wall x 2.25 blockwork with the future door reveal. The FF slab and the roof over the shaft (2.25 × 2.2) are **framed as openings now and closed with removable precast / lightweight panels** on hidden beams (B11 and the x 2.25 line), so the shaft opens later without cutting. Pit 1.2 m, isolated from the air tunnel.
- **Air tower** (8.95–10.594 × 0–1.0): rises through both slabs; the openings framed by B15 / R9 (y 1.0) and the walls x 8.95 (W3) and the cut wall (W4); the head above the roof a light RC frame.
- **Roof openings:** courtyard sky framed by R1, R2, R3 (west, cantilever strip) and R4; suite court by R5, R6 and the FF walls x 5.6 / x 11.586 below the roof. West shaft (1.5 × 1.2) framed in B13 / R10.

Conflicts and open questions for the structural engineer

No.	State	Conflict / question
X1	accepted	Column **C22** (5.6, 13.0) stands in the courtyard / living glazing (x 4.45–8.75): the glass is split 4.45–5.45 + 5.75–8.75. A decision of the columns solution (tag v5.0); keep the column round Ø0.30 and detail the glazing around it.
X2	accepted	Beam **B5** (x 8.95, 6.25 m, 0.25 × 0.55) shows 0.35 m below the dining ceiling, on the ceiling-height change line, soffit ≈ +2.85. Acceptable: it is the step between the two heights. Alternative 0.60 × 0.40 (soffit +3.00).
X3	to fix	Column **C33** (13.311, 10.45) clashes with the first-floor suite bath east window (y 9.6–10.35): move the column to y 10.6 (the FF bath / dressing wall is on y 10.45, a 0.15 m offset is hidden in the wall) or shorten the window to y 9.6–10.2. The GF kitchen window (y 8.7–10.1) is clear.
X4	rule	The gym roof's north edge **R6** (y 21.25) must be a **drop** beam, not upturned: the gym roof edge is a railing, not a parapet (solid top +7.20). An upstand of 0.40 would raise the solid top to +7.60 and cut the winter noon sun on the clerestory windows from ≈ 81 % to ≈ 40 %.
X5	flag	**R6** (6.0 m) is carried by the first-floor walls x 5.6 and x 11.586, which stand on thickened ribs of the men's majlis roof (B20, B21), not on columns: a transfer (wall on a beam). Reason: no column may stand inside the men's majlis. Build those two walls as RC walls 0.20 m (2 m long, from the slab to the roof) so the path roof → wall → rib → GF walls is explicit; the engineer sizes B20 / B21 for it.
X6	flag	The gym's north wall **y 21.25** (7.7 m) runs across the majlis roof ribs with nothing below: a **light partition** is recommended (100 mm block or dry wall; high obscured windows onto the court). If the owner wants a 200 mm wall, a solid band in the ribbed slab. Gym live load 5 kN/m² and the treadmill's vibration: natural frequency of the 4.75 m ribbed slab to check; floating floor and pads in the arm (DETAILS.md).
X7	flag	The living's south wall (y 19.25, x 5.6–8.95) is 7 m tall with the clerestory windows, tied at +3.6 only on its south face (men's majlis roof slab) and at +7.0 by the roof: fine for a block wall between columns C23 and C28 (3.35 m), but the engineer confirms the slenderness; RC stiffeners at the window jambs if needed.
X8	flag	**Colonnade columns C21–C23** end at the first floor (they carry only the edge beam B3). The roof over the gallery strip (1.35 m, y 6.5–13.0) then cantilevers from R3. Option: continue C21 and C22 to the roof inside the gallery railing line (two columns in the gallery, a two-storey colonnade) with a roof edge beam on x 5.6; cleaner structure, two more columns in the upper gallery. Owner / engineer.
X9	missing	**No soil report:** footing type and depth (pads ≈ 1.5 × 1.5 assumed; Taif granite / weathered rock is usually good), the lift pit (1.2 m), and the **air tunnel under the east band** crossing the ground beams: tunnel depth and the footings of C24–C27 / W3 to coordinate.
X10	flag	**Suite walls over the kitchen** (x 10.3, y 10.45, x 12.211, y 8.0, y 11.9, y 13.9): light partitions on the ribbed slab, or hidden beams under the bath walls that carry pipes; the **suite bath floor is sunken** (depressed slab ≈ 0.15 over x 10.3–13.311, y 6.5–10.45) with nothing in the kitchen ceiling: a depressed-slab detail inside a ribbed bay.
X11	flag	**Floor-to-floor 3.60** with a 0.20 slab and 0.50–0.55 drop beams gives 2.85–2.90 m clear under the beams (B4 over the riwaq glazing, B5 at the dining step, B1 in the women's majlis if dropped). Acceptable (≥ 2.40), but the owner should see the women's majlis option (B1 hidden in a 0.30 ribbed slab keeps a flat 3.30 m ceiling).
X12	open	**Beam B4** (x 4.25, 6.5 + 6.25 m) carries the first-floor bedrooms' east wall over the riwaq glazing (y 6.8–8.9, 10.7–12.7) and the low wall: with 0.25 × 0.50 the glazing heads are at ≤ +2.90; confirm the riwaq window heights with the architect.
X13	open	**Which walls are RC:** this study proposes W1–W5 (lift, shaft, laundry wall, cut wall, kitchen north wall) as RC 0.20 m for stiffness, plus the two gym walls of X5. The engineer confirms the lateral system (frame + walls) under SBC 301 and the wind / seismic case for Taif.
X14	open	**Slab types:** ribbed 0.30 (hollow block) for the 4.36–4.75 m bays and the men's majlis roof; solid 0.20 elsewhere and 0.22 flat over the double-height living. The engineer may unify (all ribbed, or all solid 0.20–0.25) for buildability; the clean soffit over the void is the only architectural wish.
X15	open	**Stair:** flight widths (1.375 m assumed in a 2.75 m well), the L-shaped opening and the trimmers B16–B18, the store under the half landing (≈ 1.6 m headroom), handrails; confirm the arrival strip y 6.2–7.9 at the first floor.

Rules of thumb used (all assumed): drop beams depth ≈ span / 12, hidden beams = slab depth, solid one-way slabs ≈ span / 24–28 (0.20 for ≤ 5 m), hollow-block ribbed slabs 0.30 (25 + 5) for 4–6 m, flat solid roof 0.22 for 4.7 m, columns 0.25 × 0.50–0.60 in the walls (two storeys, light loads), colonnade Ø0.30, pad footings 1.5 × 1.5 m. Loads: SBC 301 (residential 2 kN/m², gym 5 kN/m², roof 1 kN/m² + plant). Nothing here replaces the structural design.

✓ **Preliminary structural study · 36 columns · 21 first-floor beams · 10 roof beams · 15 items for the engineer**